

RONNEBY, Sweden

WMO: 026640

Lat: 56.267N

Lon: 15.268E

Elev: 190

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	14.5	5.1	7.2	11.9	10.1	9.3	17.6	24.3	38.9	21.7	37.4	4.8	10	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	79.9	65.9	76.7	64.6	73.6	63.3	68.7	75.6	66.7	73.2	64.9	71.2	9.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.2	97.2	71.7	64.3	90.7	69.9	62.5	85.1	68.0	32.9	75.7	31.3	72.9	30.0	71.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.7	15.7	1.8	84.8	6.8	3.6	-3.0	87.4	-7.0	89.4	-10.8	91.4	-15.7	94.0		
(5)			1.3	71.5	6.8	2.1	-3.5	73.1	-7.5	74.3	-11.3	75.5	-16.2	77.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	31.1	31.6	35.4	43.0	51.1	57.7	62.5	61.7	54.8	46.5	39.2	33.5	(6)	(7)
	DBStd	12.89	7.63	7.32	6.74	6.17	6.27	5.12	4.92	4.74	5.06	6.11	6.53	8.18	(7)	(8)
	HDD50	2838	585	516	455	224	61	4	0	0	16	140	326	511	(8)	(9)
	HDD65	7087	1050	936	919	659	433	226	110	123	306	574	775	976	(9)	(10)
	CDD50	1288	0	0	0	15	94	235	388	364	160	31	1	0	(10)	(11)
	CDD65	64	0	0	0	0	1	7	33	22	1	0	0	0	(11)	(12)
	CDH74	589	0	0	0	1	19	85	324	157	3	0	0	0	(12)	(13)
	CDH80	79	0	0	0	0	1	10	52	16	0	0	0	0	(13)	(14)
(14) Wind	WSAvg	7.3	7.9	7.9	7.8	7.6	7.5	7.3	6.7	6.5	6.9	7.1	7.2	7.3	(14)	(15)
(15) Precipitation	PrecAvg	26.4	2.0	1.7	1.2	1.5	2.1	2.4	2.6	2.6	2.4	2.6	2.8	2.4	(15)	(16)
	PrecMax	33.0	3.3	3.3	2.8	3.0	5.2	6.3	5.5	6.0	7.2	5.9	5.9	6.3	(16)	(17)
	PrecMin	19.2	0.0	0.6	0.2	0.1	0.4	0.0	0.2	0.9	0.5	0.2	0.5	1.0	(17)	(18)
	PrecStd	3.2	0.9	0.9	0.7	0.7	1.0	1.2	1.2	1.4	1.8	1.4	1.3	1.1	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	48.5	57.8	70.0	76.5	80.8	84.7	82.1	73.3	62.5	54.7	50.1	(19)	(20)
		MCWB	45.7	44.8	47.7	56.5	63.4	65.2	67.3	67.4	64.2	57.6	52.6	48.9	(20)	(21)
	2%	DB	44.5	46.0	53.2	63.3	72.0	76.2	81.0	78.4	69.7	59.1	52.0	47.8	(21)	(22)
		MCWB	43.3	43.7	45.6	52.1	59.6	63.4	66.6	65.1	61.6	56.1	50.4	46.5	(22)	(23)
	5%	DB	42.5	43.5	48.4	59.0	68.0	72.4	77.7	75.1	66.6	57.1	49.8	45.9	(23)	(24)
		MCWB	41.3	41.5	42.9	49.8	57.0	61.3	65.3	64.9	59.9	54.8	48.5	44.8	(24)	(25)
	10%	DB	40.7	41.2	44.9	55.2	64.2	69.4	74.5	71.9	63.9	55.4	48.1	43.9	(25)	(26)
		MCWB	39.7	39.6	41.1	47.3	54.7	59.5	63.9	63.8	58.5	53.3	47.1	42.9	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	46.5	50.0	57.8	65.2	68.7	71.4	70.8	66.2	59.2	53.3	49.3	(27)	(28)
		MCDB	47.0	47.1	55.5	67.8	74.5	77.5	80.0	76.5	71.1	61.2	54.1	49.8	(28)	(29)
	2%	WB	43.5	44.0	46.3	53.6	61.3	65.2	69.1	68.4	63.0	57.0	50.8	46.7	(29)	(30)
		MCDB	44.1	45.2	51.3	61.5	68.7	73.2	77.1	73.8	67.3	58.5	51.7	47.3	(30)	(31)
	5%	WB	41.4	41.7	43.7	50.9	58.6	62.7	67.1	66.4	61.1	55.3	48.6	44.8	(31)	(32)
		MCDB	42.0	42.9	47.3	57.3	65.7	69.8	74.5	71.6	64.8	56.7	49.3	45.5	(32)	(33)
	10%	WB	39.6	39.7	41.6	48.3	55.9	60.6	65.2	64.8	59.3	53.8	47.1	42.9	(33)	(34)
		MCDB	40.3	41.2	44.4	54.2	62.2	67.3	72.1	70.1	62.6	55.2	47.8	43.6	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.9	14.1	18.4	19.3	18.4	18.0	17.2	15.6	12.2	9.0	8.4	(35)	(36)
		MCDBR	7.9	11.1	20.4	25.2	25.2	23.9	24.4	21.7	19.1	13.3	8.7	8.2	(36)	(37)
	5% WB	MCWBR	7.4	9.0	13.7	15.7	14.6	13.3	12.1	11.6	12.1	10.3	8.1	8.0	(37)	(38)
		MCWBR	7.2	9.8	17.9	22.6	21.7	21.3	20.6	18.4	16.7	12.1	8.4	8.0	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.306	0.333	0.351	0.384	0.382	0.374	0.398	0.397	0.367	0.355	0.335	0.298	(39)	(40)
		taud	2.289	2.319	2.343	2.320	2.383	2.434	2.385	2.388	2.452	2.402	2.327	2.256	(40)	(41)
	Ebn at Noon		187	229	257	264	272	276	266	258	250	219	174	157	(41)	(42)
		Edh at Noon	17	24	30	35	35	34	35	32	27	22	16	13	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		132	323	778	1296	1703	1807	1652	1336	930	440	168	94	(43)	(44)
	RadStd		24	58	76	160	172	121	198	118	108	77	32	9	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+3.38	N/A	N/A	N/A	N/A	-256	N/A	N/A
(47)	Regional (46 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-292	N/A	N/A

Nomenclature: See separate page